TECHNOLOGICAL UNIVERSITY (THANLYIN)

DEPARTMENT OF INFORMATION TECHNOLOGY ENGINEERING

OBSTACLE DETECTION FOR RASPBERRY PI BASED RESTAURANT SERVING ROBOT

BY

MA INKYINN THAW THAW

VI IT - 2

GRADUATION THESIS

OCTOBER, 2025

THANLYIN

TECHNOLOGICAL UNIVERSITY (THANLYIN)

DEPARTMENT OF INFORMATION TECHNOLOGY ENGINEERING

OBSTACLE DETECTION FOR RASPBERRY PI BASED RESTAURANT SERVING ROBOT

BY

MA INKYINN THAW THAW

VI IT - 2

GRADUATION THESIS

OCTOBER, 2025

THANLYIN

TECHNOLOGICAL UNIVERSITY (THANLYIN)

DEPARTMENT OF INFORMATION TECHNOLOGY ENGINEERING

OBSTACLE DETECTION FOR RASPBERRY PI BASED RESTAURANT SERVING ROBOT

Ma Inkyinn Thaw Thaw

VI IT - 2

A THESIS
SUBMITTED TO THE DEPARTMENT OF
INFORMATION TECHNOLOGY ENGINEERING
IN PARTIAL FULFILMENT OF THE REQUIREMENTS
FOR THE DEGREE OF BACHELOR OF ENGINEERING
(INFORMATION TECHNOLOGY ENGINEERING)

OCTOBER, 2025

THANLYIN

TECHNOLOGICAL UNIVERSITY (THANLYIN)

DEPARTMENT OF INFORMATION TECHNOLOGY ENGINEERING

We certify that we have examined, and recommend to the University Steering Committee for Bachelor of engineering Graduation thesis entitled: **“OBSTACLE DETECTION FOR RASPBERRY PI BASED RESTAURANT SERVICING ROBOT”** submitted by **MA INKYINN THAW THAW**, Roll No. **VI-IT 2 (November, 2024)** to the Department of Information Technology Engineering as the requirement for the Bachelor of Engineering degree **B.E (IT)**.

Board of Examiners:

1. Dr. Mon Mon Zaw
Professor and Head
Department of Information Technology Engineering ....
Technological University (Thanlyin) (Chairman)

2. Daw Myat Thida Soe
Associate Professor
Department of Information Technology Engineering ....
Technological University (Thanlyin) (Supervisor)

3. Daw Nge
Associate Professor
Department of Information Technology Engineering ....
Technological University (Thanlyin) (Member)

4. Daw Thu Thu Aung
Associate Professor
Department of Information Technology Engineering ....
Technological University (Thanlyin) (Member)

5. Daw Swe Thiri Maw
Demonstrator
Department of Information Technology Engineering ....
Technological University (Thanlyin) (Member)

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who have supported me in the completion of this thesis. First and foremost, I am deeply grateful to Dr. Theingi who is the Rector of Technological University (Thanlyin) for creating an inspiring academic environment and for the guidance and encouragement that have shaped my scholarly journey.

I would also like to extend my heartfelt thanks to Dr. Mon Mon Zaw who is the Head of the Department of Information Technology Engineering for her invaluable advice, encouragement, and unwavering support throughout this thesis.

My deepest appreciation goes to my supervisor, Daw Myat Thida Soe, for her expert guidance, constructive feedback, and patient mentorship. Her insight, knowledge, and encouragement have been instrumental in the development and completion of this thesis.

I am also indebted to my teachers and colleagues in the department for their kindness, knowledge, and encouragement. Their thoughtful discussions and assistance have enriched my understanding and contributed greatly to this thesis.

I owe a special debt of gratitude to my parents whose steadfast love, encouragement, and belief in me have been my greatest source of strength. Their sacrifices and constant support made this achievement possible.

ABSTRACT

This thesis presents the design and implementation of an obstacle detection system integrated with line-following functionality for a restaurant serving robot using a Raspberry Pi. The robot autonomously navigates by following a black line on the floor using infrared sensors that detect line contrast. To ensure safe operation in dynamic environments, a Time-of-Flight (TOF) sensor is used to detect obstacles ahead. When an obstacle is identified within a defined range, the robot halts its movement and simultaneously issues a voice alert through connected speakers to notify nearby users. The Raspberry Pi serves as the central processing unit, continuously collecting sensor data, processing information in real time, and executing motor control commands alongside alert activation. The software, developed in Python, employs threshold-based decision-making to maintain accurate line tracking while responding promptly to obstacles. This straightforward yet effective method avoids the complexity of rerouting algorithms, making the system well-suited for structured indoor environments such as restaurants. Experimental testing demonstrates that the robot reliably follows its designated path and detects obstacles with accuracy, stopping and alerting as intended. The voice alert mechanism enhances user interaction and safety, while the modular system design allows for potential future improvements, such as rerouting functions or integration of more advanced sensors. Overall, this work contributes to the advancement of service robots in hospitality by combining reliable navigation with efficient obstacle detection and interactive voice feedback.

TABLE OF CONTENTS